Observer Patch Holography

A Compact Case for an Observer-First Unification Program

Paper release: r2023 Released: August 20, 2026

Abstract

Observer Patch Holography (OPH) starts with finite self-reading systems that expose boundaries, keep records, compare overlaps, and repair disagreement. From that one architecture: public records form an exact event algebra with Born probabilities, Lüders conditioning, and the Tsirelson bound; a conditional finite theorem package yields the four laws of thermodynamics once its named source receipts are supplied; Lorentz kinematics holds on the spherical screen and typed modular, stress, and entropy premises compose to the Einstein relation; a classification theorem forces the complete twelve-port response to have the Standard Model gauge Lie type without a chosen ambient group; and one exterior branching returns a fifteen-state generation with cancelled anomalies. Deterministic simulations measure a Lorentzian event form with one time and three space directions across a support-adjusted three-rung path; a preregistered fresh-seed replication reproduces it at the middle rung on five of five replicates and demotes the small-rung and attribution readings under its frozen rule. Interval arithmetic certifies unique roots of the declared pixel maps, a positive-chamber circulant theorem gives the Koide relation exactly, and finite capacity identities supply a conditional de Sitter time-advance mechanism. A prospective primitive-port linked-coefficient test predicts three exact dispersion coefficients and a unique degree-six icosahedral angular shape under custody-bound targets, kill bands, and decision rules fixed before comparison. Every result states its premises, evidence class, and physical boundary in place; the open physical attachments are explicit. A sorry-free Lean library, exact code receipts, and reproducible simulations make the case testable at each step.

OPH asks whether physics can be reconstructed from the minimum machinery required for a public world. A patch has local state, a boundary, records, readback, and repair moves. Each patch sees a fragment. Agreement across shared boundaries turns private descriptions into public facts. The selected world is a stable normal form,

$$T(\mathfrak{U}) = \mathfrak{U}.$$

The theory rests on three core axioms.

The three axioms

1. Oriented twelve-port observer screen. There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron; carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Formally: for every regulator r there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$ whose carriers carry twelve primitive central port projections and the exact boundary packet $K = (P, E, F, o)$, joined by seam algebras with coherent triple-overlap cocycles into a nerve N_r with a degree-one bridge b_r to the oriented spherical support S_r, all commuting with refinement. A complete carrier has a finite-dimensional unitary reversible response space and an injective real-linear port derivative $D_r : \mathbb{R}^{P_r} \to \mathfrak{u}(H_r)$ with positive trace pairing, commutator-closed image, and refinement-natural quotient-visible response.

2. Observer agreement. Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally: the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map on accepted public data. Accepted reversible overlaps form a groupoid surjecting onto the proper carrier automorphisms. For each such automorphism, one closed path has a projective implementer whose adjoint action is covariant on D_r and lies in the response-generated identity component modulo the pointwise centralizer.

3. Conditional maximum randomness. Everything that observer agreement leaves unconstrained is maximally random. Formally: the realized state is the information projection of an exact reference family onto the convex set of compatible local state families satisfying the finite observer-visible constraints. The finite A1-generated observer cover is state-determining on that feasible set, and its exact weights are strictly positive: $\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_P w_{r,P} D(\rho_{r,P} \| \tau_{r,P})$.

The first axiom constrains architecture, the second meaning, the third state selection. None of them contains a gauge group, a particle list, a recovery law, or a rule that fixes field content or multiplicity. Every result below states its axiom dependencies, further premises, mathematical domain, and physical boundary. The complete plain and formal statements, scope clauses, and caveats are collected in the canonical axiom reference.

One global closure principle sits beside the axioms without joining them. The framework models the universe as a self-referential fixed point: the simulating and the simulated description are one system. Once independently constructed outer and inner readings are proved to denote one invariant, self-identity forces their equality. The physical bridge and return map are premises of that reading; existence, uniqueness, and stability are separate tests. Terminology does not change the equality: unequal readings after the typed identification would describe two systems. The local screen-grain map $P = \varphi + \sqrt{\pi}/A_T(P)$ has one interval-certified root on its declared domain. Its physical interpretation requires the hadronic attachment, target-independent map selection, and the typed same-quantity bridge. For direct capacity closure, the exact all-rung arithmetic proves nonidentifiability on the bounded completion class defined by base agreement, positivity, and the carrier bound. Universal all-rung membership in a complete A1–A3 capacity-source contract and an executable-to-Lean bridge are absent. Direct N is therefore not evaluable on its incomplete source antecedent. The stronger source-class verdict is unproved. A separate common-load branch evaluated at the independently constructed source-forward coordinate P_{fwd} defines $N_0 = \pi \exp[6\pi/(P_{\text{fwd}} \alpha_U(P_{\text{fwd}}))]$, where $\alpha_U(P_{\text{fwd}})$ is the unified-coupling coordinate selected by the same forward pixel-closure solve. On the finite collar branch, declared total reserve expectation $P_{\text{fwd}}/4$ and six-class equidistribution give presence probability $P_{\text{fwd}}/24$ for each class. The source-typed candidate is $N_{\text{pres}} = N_0(1 - P_{\text{fwd}}/24)$ if one class is physically selected, its scalar-weighted receipt is discharged, and its collar-survival factor is proved to act on global capacity. The alternative $N_0 e^{-P_{\text{fwd}}/24}$ needs a mean-count or continuum carrier. The finite datum does not select either global action. Exact neutral and multiplicative completions on the same abstract cut-count source are positive and compose under disconnected cuts and finite regrouping, but disagree after one cut. The same datum selects neither a one-class blocked event, a six-class-total event, nor no capacity action. The named-law branch therefore has no global-capacity object for a horizon identification. A positive construction requires a stronger source-derived global action. The two expressions retain their exact conditional arithmetic, and their respective 0.63 and 0.39 percent offsets from the Planck base-ΛCDM comparison coordinate are retrospective and carry no predictive weight. On the source-forward branch the three numerical coordinates are

$$N_0 = 3.5321315434\ldots \times 10^{122}, \quad N_{\mathrm{pres}} = 3.2920978773\ldots \times 10^{122}, \quad N_{\mathrm{Pois}} = 3.3000722254\ldots \times 10^{122},$$

against the weighted Planck comparison coordinate $3.3129270981\ldots \times 10^{122}$.

The proposal earns attention because several familiar structures arise from this small basis. They also arrive with different evidence types. Exact theorems, interval certificates, conditional physical compositions, and simulation measurements occupy separate evidence classes throughout this paper.

The claim in one sentence
Finite observer-like patches provide a common construction for public quantum records, Lorentz kinematics, a conditional Einstein branch, compact gauge structure, matter representations, and quantitative closure tests. The mathematics fixes a substantial skeleton. Physical carrier identifications determine how much of that skeleton describes our universe.

1. Thirteen receipts that carry the case

The following count contains the strongest independent pieces of evidence. Every row states its boundary in the same breath as its result.

	Receipt	Result and boundary	Evidence
1	Public records and finite event algebra	Terminating repair gives protected normal forms on the declared confluent carrier class. On a separately declared finite algebra-state representation, the public projector span is exactly star-isomorphic to functions on its nonzero record labels, and the projectors obey Born probabilities, Lüders conditioning, and the Tsirelson bound. A common isometric copier can sharply copy two distinct alternatives only when they are orthogonal. The general mixed-state no-broadcasting implication remains a separate adapter. Consensus does not derive the quantum representation. On the twelve-port carrier, atomic signed record events generate integer loads and conservative repairs terminate by exact quadratic descent.	Lean and exact finite code [1,2,10]

	Receipt	Result and boundary	Evidence
2	Four laws of thermodynamics from conditional repair	Axiom 3 on states gives the Gibbs family by the exact information-projection Pythagorean identity. On transition distributions over the repaired visible fibre it gives weighted conditional resampling from the same reference, the observation-fibre projector of the consensus construction. That kernel is stochastic, idempotent, reversible, stationary, and fixes fibre-measurable charges, and relative entropy to the reference contracts under it: the second law is data processing for repair, with the modular form $\Delta S \geq \Delta\langle K\rangle$ and the Landauer bound as corollaries. On one nondegenerate finite spectrum, equality of Gibbs distributions identifies the inverse temperature; no contact dynamics is proved. For a finite joint update the exact split is $\Delta U = \delta Q + \delta W + \operatorname{Re}\operatorname{tr}(\delta\rho\,\delta H)$; the two-term form is first-order or exact for an ordered fixed-H or fixed-ρ stroke. This is a bookkeeping realization, not a source-derived conservation law, and the excited-mass bound $\frac{d-g_0}{g_0}e^{-\beta\Delta}$ gives, as a standard finite corollary, entropy limit $\log g_0$ with finite-step unattainability. The strict-descent normalizer that settles public facts is a different map and carries no entropy inequality. Faithful stationarity alone also implies relative-entropy contraction; an exact lazy directed three-cycle separates that result from detailed balance. The global refinement theorem uses one supplied ambient-cofinal spectral family with no maximal regulator and strict finite-carrier growth; the same family carries uniform/cofinal-tail concentration. A supplied calibrated-stage energy identification yields the memberwise mass equality, and an explicit unbounded ladder excludes constant-carrier false greens. No source or physical continuum identification is inferred. The global objective representation, common source-derived reference, source collar realization, energy-clock calibration, and refinement-uniform low-temperature control are premises. Exhausting the fifteen nonempty field-subset projections of the pinned source table yields only a nonreversible eight-state probe and a singleton fine recurrent class. The state table/reference was locally pre-specified and hash-pinned, while the recurrent transition chain was extracted post hoc from retained run data. They have two exact direct-binding obstructions: the state-side resampling action is idempotent, whereas the transition action has a nonconstant eigenmode with $$\lambda = \frac{665437}{726948} \in (0,1).$$ Every intertwiner into an idempotent action therefore kills that mode. In addition, the chain stationary mass 7155/61511 lies strictly between 1905/16384 and 1906/16384, so it is not a deterministic pushforward of the 16384-sample empirical reference. These results exclude the two audited direct-binding mechanisms on the current source artifact. The committed random-scan preflight decides the random-scan instance within its certified grammar: in the two certified arenas, every computed uniform-scheduler mixture of conditional-resampling projectors escapes the idempotent obstruction, while under both declared schedulers every computed mixture has a one-dimensional fixed space. The analytically covered constant-field cases carry only the fixed-space conclusion, so that grammar protects only constants, and the recorded continuations are an enriched export or a dilated construction. Nonlinear, reverse-direction, and enriched-source routes stay open, and none of this discharges the physical premises of the conditional package.	Lean and exact rational code [1,9]

	Receipt	Result and boundary	Evidence
3	Forced dynamics, forced Born weights, and a derived action	On the finite private algebra every pointwise-continuous symmetry flow is blockwise unitary conjugation generated by one time-independent self-adjoint Hamiltonian per block, unique up to a real scalar: the Schrödinger form is a theorem, and its single continuity premise is displayed. On the same finite surface every nonnegative normalized valuation additive on coexisting effects is the trace against one density matrix, in every finite dimension including two, so Born weights are derived without a continuity axiom; sharp two-outcome webs admit an explicit non-Born valuation. One unsharp three-effect context excludes that valuation, but the declared finite battery admits a different planar cubic response and therefore does not supply the full coexistent-effect additivity hypothesis. The phase lift obtained by subtracting the normalized complex commutator of two source-attached algebraic projector candidates from $I/2$ is exactly the missing Pauli-Y projector and completes fixed-trace operator tomography, while every effect in a generous phase-free real/Kraus closure remains Pauli-Y blind. This is an algebraic closure theorem. A declared Lüders instrument for the committed effects supplies completely positive outcome maps and a trace-preserving summed channel per context, and the effect table does not determine the instrument; the source supplies neither the instrument selection nor its outcome receipts. The path law of the realized source chain is the exponential tilt of its declared reference by the log-transition action, unique up to an additive constant and a multiplier rescaling: the action is derived, not chosen, and the committed repair-count action reproduces the law at no multiplier. A separate finite Legendre bridge makes the discrete Euler–Lagrange condition equivalent to one discrete Hamilton step for a supplied regular Lagrangian and supplies quadratic momentum, energy, and Noether witnesses. The realized-chain probability/action theorem and that Legendre witness are conjoined in the composite receipt; no theorem produces the realized chain from the Hamilton flow. More strongly, the binary history law does not select a real Legendre enrichment: its bilinear extension is velocity-affine, whereas adding $a\,y(y-1)/2$ leaves every realized history unchanged and, for each $a>0$, gives a regular strictly convex Lagrangian and Hamiltonian; the checked $a=1,2$ members are distinct. A declared mode-extremality rule fixes one member only inside that one-parameter ansatz: the exact family $L_{a,c}=L_0+ay(y-1)/2+cx(x-1)/2$ preserves every binary corner and the same constant-history stationarity and fixed-endpoint single-site variation minimum whenever $a+c=a_\star$, while its velocity curvature varies with a. A velocity-only clause selects the committed point only inside that quadratic grammar. It does not select globally: for $0<\lambda<a_\star$, the quartic continuation $L_{a_\star}+\lambda y^2(y-1)^2$ preserves every binary corner, the same junction stationarity, the fixed-endpoint single-site minimum, and a globally invertible momentum map while differing from every quadratic. Conditional resampling with a declared counting reference and trivial visible datum realizes the uniform transition kernel, but does not derive those inputs, the initial law, or the complete path reference from the source. At the separately declared empirical mean, the matching positive exponential parameter exists uniquely; this identifies a parameter conditional on a target, not the target itself. A separate concave-action control shows only that positive Gibbs weighting need not make every stationary history a mode or minimizer; it leaves constrained saddles, complex stationary phase, and source-selected real enrichments open. Thus the declared bundle is compositionally inhabited; source selection and the physical least-action interface are registered premise work. A declared equal-weight rule on the twenty oriented faces gives exactly $B_{\text{face}}=60R_{13}$, but 240 Jacobi coordinates fail. Under each explicitly added upper-triangular coordinate norm $\ell_1,\ell_2,\ell_\infty$, exact primal–dual comparison with the certified compact locus uniquely favors family G. The respective G,F,P distances are $30(\sqrt{5}-1),60,60$ for ℓ_1; $(615-123\sqrt{5})/22,(615+123\sqrt{5})/22,45$ for squared ℓ_2; and $(5-\sqrt{5})/10,\sqrt{5}/5,1/2$ for ℓ_∞. The F ℓ_1 value is an unattained infimum. The quadratic verdict extends to the complete cone of invariant carrier metrics: P is excluded everywhere, every sector-balanced metric and the whole box $\beta/\delta\in[1/50,6]$ select G, while the mirror family F occupies a nonempty region across the exact three-scale tie surface. The committed uniform port-dual measure (1/12 per port from equal face weights and the barycentric share) induces exactly the balanced cone point, where G is uniquely nearest at twelve times the reference distances. The weighting rules, measure-to-metric identification, nearest-point rule, and metric class are not source-derived, so this is a measure-conditional discriminator, not bracket selection. For the supplied brackets, ad-invariance reduces the F/G carrier-form space from three weights to two: F imposes $w_{3+}=\sqrt{5}w_5$, G imposes $w_{3-}=\sqrt{5}w_5$; the P control remains two-to-two. A symbolic two-factor theorem binds constructed Gibbs kernels for both P factors and the full color-bearing F family. Equality of such kernels leaves only common scaling, so the relative coefficient is identifiable but not selected. No independent source kernel, physical unit, clock, or continuum field is constructed.	Lean and exact kernel receipts [9,15]

	Receipt	Result and boundary	Evidence
4	A continuous local three-dimensional carrier and Lorentz kinematics	On the certified spherical support branch, $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ and $\dim[\mathrm{SO}^+(3,1)/\mathrm{SO}(3)] = 3$. Event population is a separate physical receipt. On a distinct conditional branch, the declared discrete repair mean and the normalized equal-port response limit select a rank-three spectral projector and its Gram form. The limiting form has a three-dimensional radical on the real six-control extension. The integer signed-record module has no metric kernel, embeds densely in the resulting quotient, and completes to an abstract continuous local Euclidean translation carrier. Finite-step completion gives a different object. Selection of the repair law and readback topology, isometric carrier action, overlap gluing, scale, and physical-space identification are open.	Theorem chain, Lean, exact code [1,3,10]
5	Einstein composition and a finite signature test	The typed modular, null-stress, entropy, and small-ball premises compose to the Einstein relation. On a declared chart with one ancestry coordinate and three spectral coordinates, a support-adjusted three-rung path measures inertia $(1, 3)$ with cone margins $-5.62, -3.22, -1.41$. A same-size support-width control changes the inertia to $(2, 2)$. A preregistered fresh-seed replication reproduces $(1, 3)$ at the 65,536-carrier rung on five of five replicates and returns FAILED overall: the 16,384-carrier rung is seed-fragile, the declared margin-ratio band is missed, and the ancestry-permutation null does not discriminate, so the emergent reading is demoted with the receipts on the instrument record. The instrument tests signature and does not select the chart dimension. One target-clean finite source also inhabits exact causal, atlas, seam-transport, and local operator domains. Its six local fits include one $(0, 4)$ chart, and its cone margins are negative.	Lean implication, instruments, simulation [3,11]
6	Standard Model local gauge algebra without a chosen gauge group	The complete reversible port response in A1 and endogenous overlap transport in A2 imply an injective compact commutator-closed twelve-dimensional current with inner A_5 action. Its one-dimensional fixed space and compact classification force $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. No ambient Lie group, factor dimension, or matrix current is inserted. The finite-source simulator derives the carrier module and $R = -J$. The source contains no ordered current tomography or same-current holonomy; the current identification consumes both as premises.	Lean, compact classification, exact controls [3,10]
7	Global quotient and one-generation representation witness	For the declared tensor representation, trace balance gives $S(U(3) \times U(2))$, the common action kernel is $\mathbb{Z}_6$, and an exterior algebra branches into one fifteen-state Standard Model generation with the anomaly arithmetic. The physical global quotient requires a source-derived complete character category and a same-source loop-to-kernel identity. Among the single complete faithful screen bands in $3 \le N_g \le 5$, the exact Laplacian cost selects rank three, and the declared unitary simulator recovers that band at its lowest positive generator frequency. Tensoring the band with the declared generation table gives a conditional complex rank-45 candidate. The chirality grading and diagonal $\mathbb{Z}_6$ action belong to that table. A separate local-domain receipt checks the declared extension $D_\sigma \otimes I_{45}$ and conditional inheritance of its positive finite gap. The source does not select this matter action. The twelve-port Spin packet and the 8,662-node local domain have no certified source, domain, or transport bridge. Matter-pole identification, continuum Spin/locality, physical seam selection, and laboratory attachment are premises of any physical reading.	Lean and exact code [3,10]
8	Strict W/Z analytic checks	A strict one-loop consumer maps a complete renormalized packet to charged and neutral scalar pole coordinates. Interval receipts exclude zeros on the declared principal-sheet boxes and isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet and prove no unique continuation, sign bridge, full-matrix Laurent residue, physical current amplitude, or independent numerical replay. The external fixture is not composed with the OPH electroweak chart.	Conditional analytic receipts, exact code [1,6,9]
9	Pixel fixed-point arithmetic	Outward-rounded interval certificates prove existence and uniqueness for each declared pixel map and exclude a second root across its analytic domain. The fine-structure comparison is diagnostic because the physical transport map, map selection, and same-quantity bridge are incomplete.	Interval arithmetic [5,10]
10	Charged-lepton interval diagnostic	The measured electron, muon, and tau masses lie inside outward-rounded transport enclosures. Their logarithmic half-width is 1.732%, giving one-sided multiplicative widths of −1.72% and +1.75%. Target-anchored ratios and empirical transport make this a closure diagnostic.	Interval certificate [6,10]
11	Positive-chamber Koide theorem	A Hermitian three-cycle response obeys $Q = \frac{1}{3} + \frac{2}{3}(\lvert b\rvert/a)^2$. Hence $Q = 2/3$ exactly at $\lvert b\rvert/a = 1/\sqrt{2}$. Equal finite event blocks supply this balance under the declared tracial packet. Under the balance and ordering premises the measured electron and muon masses fix the tau mass inside a 72-eV enclosure, 0.43σ from measurement. The conditional test has a fixed precision floor and rejection rule. No phase selection or physical family attachment is claimed.	Lean and exact code [7,9,13]

	Receipt	Result and boundary	Evidence
12	Finite de Sitter identities	Pure de Sitter normalization, the finite entropy maximum, the exact capacity transfer law, its analytic curvature, and the port/edge line-graph identity are exact. The gravitational shock sign requires the stated horizon and observer dictionaries.	Lean and exact code [8,10]
13	Frozen carrier-ray propagation predictions	On the real reciprocal finite-range cosine branch whose sole hop support is the complete twelve-port orbit, with no independent kinetic term through the displayed order, carrier covariance forces equal weights. Continuum normalization predicts $C_4 = -a^2/20$, $B_0 = a^4/840$, and $B_6 = 2a^4/7875$, hence $B_6/C_4^2 = 32/315$, $B_0/C_4^2 = 10/21$, and $B_6/B_0 = 16/75$. Angular ranks one through five vanish and rank six has one fixed icosahedral shape up to spatial orientation. The target and branch-level rule are fixed under public commit custody. The detached timestamp evidence consists of calendar commitments and does not claim Bitcoin confirmation. The dated cosmic-microwave-background template class is excluded, and the linked coefficient relation has no qualifying comparison. The physical-sector, coherent-frame, and exclusivity bridges are premises of any branch-level verdict. The certified repair operator acts on thirty internal seams rather than spatial translations and does not select this orbit as a physical hop operator. An exhaustive classification of the simulator's declared serialized finite artifacts finds local spatial and kinetic operators on separate domains, with no registered accepted packet joining a complete twelve-port direction-orbit translation to a physical readout of that same operator. The source seam current separately fixes the edge-orbit ray $B_6/C_4^2 = -2/63$, with the opposite rank-six sign, under its own physical action premises and custody record. Its equal source-counting average has an exact Dirichlet generator on the response completion. The generator's oscillator lift is auxiliary. A basis-free transverse completion has rank two at nonzero momentum and an exact zero mode under its declared oscillator generator. The construction is a conditional free-vector model and makes no Maxwell or photon claim. Physical frequency remains a separate identification. A committed finite-seam package adds the exact rational Green matrix of the seam Laplacian, the unique minimal-energy Coulomb solution of every neutral Gauss problem, and a genuinely local face-curvature action. Its signed face map has rank nineteen and kernel exactly the port gradients; the kinetic matrix has only five nonzero entries per seam row. Gauge invariance and stationary solvability are each equivalent to conservation of the typed seam current; when a stationary point exists, its minimum is unique modulo gauge. A separate scalar action has the Green potential as its neutral minimum modulo constants, and the direct-product static theorem keeps abstract port load and seam current distinct. A declared discrete kinetic term on the same carrier completes it; the step-scaled leapfrog update and the Gauss constraint are the Euler–Lagrange equations of the resulting action, and at unit step: Faraday's law is a theorem of the staggered definitions, Gauss propagation is equivalent to the discrete continuity equation with load and current kept as distinct types, a staggered quadratic form obeys an exact source-work balance, time-dependent gauge transformations leave the fields, the evolution law, and that form invariant, static zero-current solutions with neutral port load are exactly the committed Coulomb sector, and the zero-current second-order recursion carries the committed local wave operator. For zero current the staggered form is conserved and nonnegative below the sharp threshold $h^2(3+\sqrt{5}) < 4$, where both field energies are bounded at every step; above it explicit unbounded modes exist, so the unit step is unstable. This is a conditional finite composition on a declared step index; physical time and sources, covariance, continuum control, and readout are not supplied. Exact synthetic enumeration rejects a zero leading coefficient in every replica of the declared finite stress law; its nominal 95 percent intervals cover the injected value in 3904 of 4096 replicas. The linked higher-order pair remains unresolved.	Lean arithmetic, exact code, custody records [9,13]

The accompanying Lean library contains more than 7100 theorem and lemma declarations. The count includes positive results, premise boundaries, and countermodels. It is useful because the formal statements expose assumptions that prose can otherwise hide. The finite receipts bind inputs, outputs, tolerances, and negative controls to reproducible artifacts [9].

2. Public reality and quantum records

Take a finite family of patches x_i, each with an observable boundary map. For every overlap $e = (i, j)$, the two induced records must agree. A repair move changes local state while preserving protected readout. On a terminating, frustration-free carrier with a confluent accepted repair relation, every schedule reaches the same quotient normal form. Exact code exhausts all six declared overlap schedules on the reference carrier and returns the same normal form [2,10]. The general confluence hypothesis is part of the carrier contract. Arbitrary local repair does not imply it.

The compare, write, and verify contract identifies the completed public events and their finite commutative indicator algebra. At this stage the record surface is represented inside a finite-dimensional $*$-algebra with a normalized state ρ. This algebra-state representation and its trace pairing are explicit inputs. Consensus determines which records are public; it does not derive the quantum representation. If P_E is the projector for a public event E, then

$$\Pr(E) = \mathrm{Tr}(\rho P_E), \qquad \rho \mid E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

The first expression is the Born probability on the declared algebra-state surface. The second is Lüders conditioning. For dichotomic observables in commuting wing algebras, the corresponding Clauser–Horne–Shimony–Holt (CHSH) parameter satisfies

$$|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}.$$

The bound is attained exactly: a declared maximally entangled state with two Pauli settings on one wing and two quarter-rotated settings on the other reaches $|S_{\mathrm{CHSH}}| = 2\sqrt{2}$ on the committed split interface, while every record-diagonal state on the declared split obeys the classical bound $|S_{\mathrm{CHSH}}| \leq 2$ against all slot-local unit-interval settings, so the counted classical record surface cannot attain the quantum value through any slot-local readout; cross-wing commutation alone does not carry the bound. The attaining state and settings are entirely real; the theorem isolates entanglement, non-diagonal coherence, and noncommutativity, not a necessary genuinely complex phase effect, and it does not cover arbitrary context-dependent classical instruments. On the declared binary-icosahedral spinor branch, one incidence-defined setting family realizes $|S_{\mathrm{CHSH}}| = 1 + 3/\sqrt{5}$ exactly. The exhaustive twelve-port census contains (960) maximizing quadruples, so this family is not unique. The saturating state and settings are declared data; source-selected settings and a completed-record two-wing instrument are absent. These identities belong to an audited finite event-algebra development [1,10]. The theorem applies standard finite quantum probability and conditioning to the public-record surface selected by the observer contract. It does not derive the algebra-state pair, the Born rule from pre-quantum repair records, a source-produced CHSH violation, or an arbitrary continuum quantum field theory.

3. Lorentz kinematics and the Einstein branch

An oriented conformal spherical support has the connected symmetry

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{SO}^+(3, 1).$$

The space of unit timelike observer frames is

$$H^3 \cong \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \qquad \dim H^3 = 6 - 3 = 3.$$

This is an exact kinematic result on the spherical support branch. Local icosahedral carrier geometry, the federation nerve, and the global S^2 support are distinct objects. A populated four-dimensional event base requires record separation, local charts, an open-image condition, affine transition data, a quadratic cone, and causal reachability [3].

On one common algebra-state tower, normalized modular flow supplies a local one-parameter ordering. A physical clock requires an observer-readable transition process and an explicit calibration. Half-sided modular inclusions supply positive null translations, null tomography assembles local stress, and generalized-entropy stationarity with small-ball geometry gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle.$$

The Lean development proves the typed algebraic composition. A source-derived physical tower, the continuum limit, and the physical identification maps are premises [3,10].

The simulation branch fits an event form to coordinates and causal labels produced by repair dynamics; its coefficients are not inserted into the readout. The selected path uses $(16{,}384, 128, 96)$, $(65{,}536, 256, 96)$, and $(262{,}144, 512, 384)$ for carrier count, observer count, and support width. The held-out event form has inertia $(1, 3)$ at every selected rung. Its cone margin moves through

$$-5.6, \qquad -3.2, \qquad -1.4.$$

Coupling spread decreases beside it. At 262,144 carriers, the support-width 96 control has 312 cross-observer edges and inertia $(2, 2)$; the support-width 384 row has 1,062 edges and inertia $(1, 3)$. Configurations, primary arrays, hashes, and regeneration scripts are public [10]. The data establish a reproducible support and cross-read sensitivity on the instrumented branch. They do not establish an invariant-density convergence law, an infinite-scale limit, or the missing cap-state thermal receipt.

A finite source inhabits the local operator domain

One deterministic target-clean capture at 16,384 carriers supplies a finite causal complex with 2,304 events and exact acyclicity. Its declared chart combines one ancestry-depth coordinate with three spectral embedding columns. The four columns are nondegenerate on the sample, and the global held-out form has inertia $(1, 3)$. Six closed observer-visibility neighborhoods have exact induced affine wiring transitions, cocycles, orientation, and time orientation. Five local quadratic fits have inertia $(1, 3)$; one has $(0, 4)$, and the measured cone margins are negative.

On its observer-visible seam complex, 8,662 carriers and 11,816 seams contain 38 triangles. The declared reversing seam transport frustrates all 38. Two coordinate routes through the same $\mathbb{F}_2$ rank implementation agree with the exact rank identity and give lift-ambiguity rank 3,117. Scalar, chiral, and gauge sections carry a declared unit-counting measure and a sign-twisted local derivative. Exact integer evaluations on deterministic sections test its adjoint, kinetic, covariance, gluing, refinement, and boundary relations. The signed-graph rank theorem gives a zero twisted kernel. One target-clean provenance graph binds the construction, and an isolated rerun of the same producers reproduces the canonical receipt content [10].

This finite result gives the particle, clock, and defect calculations a common typed domain. It does not supply open manifold charts, a positive cone margin, continuum Stiefel–Whitney identification, physical field selection, or a continuum operator limit.

Held-out measurement and control

The observed signature is held out from the update rule, persists across three named configurations on the retained path, replicates at the middle rung under a preregistered fresh seed while failing the frozen replication rule on small-rung stability and ancestry attribution, changes under the same-size support-width control, and sits inside a typed theorem chain that states every physical premise. This combines an exact implication with a direct computational measurement.

4. Why the complete twelve-port response forces the Standard Model gauge algebra

The declared carrier lineage realizes the twelve ports with equal algebra traces, a realization property rather than axiom content, with oriented icosahedral incidence. Its proper rotation group is the alternating group A_5. Exact finite calculations give the distance profile, the antipodal pairing, the sixty proper rotations, the rank-three Gram frame, and the decomposition of the real port-coefficient space

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

Complete reversible port response in A1 supplies an injective commutator-closed map $D : P_{12} \to \mathfrak{u}(H)$. Endogenous overlap transport in A2 implements every proper carrier recharting inside the response-generated group. The image is a compact twelve-dimensional Lie algebra with a one-dimensional A_5-fixed space. Compact classification then gives

$$\operatorname{im} D \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

The centreless alternative is $\mathfrak{su}(2)^4$. An inner A_5 action preserves every simple factor. Since A_5 is simple, each restricted action is trivial or faithful and has fixed dimension three or zero. The total cannot equal one. The surviving centre has dimension one and the semisimple part has dimensions $3 + 8$ [3,10].

Gauge structure without a chosen gauge group

The argument starts with the operational response of the twelve ports. A1 makes that response faithful, complete, unitary, and closed under sequential commutators. A2 makes every proper carrier recharting internal to the same response. Incidence supplies the A_5-module and its single fixed line, but it leaves many equivariant response laws. The response and agreement clauses remove that freedom together. The fixed-line obstruction and compact classification leave the local Standard Model gauge Lie algebra as the unique result. No ambient compact group, factor dimension, matrix current, charge table, or particle assignment is a premise. The theorem fixes the abstract local Lie algebra. Source realization, matter action, coupling normalization, the global quotient, and laboratory attachment are separate premises. A descent repair law is required for branches that consume settled public records; it is not an extra premise of this Lie-algebra theorem.

Target-blind port impulses and local recurrence readback derive $R = -J$ and its four band signs. The charged-double-triplet matrices give an exact conditional realization

$$\mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

The source contains no ordered reversible current tomography or closed overlap words tied to that same current. Matrix realization, coupling normalization, laboratory attachment, and identity with the independently reconstructed categorical current are premises of any physical identification.

The registered adjacency recurrence and inverse-port readback generate $\operatorname{span}\{I, A, A^2, A^3\}$, a commutative algebra of dimension four. Its exact polynomial closure supplies neither twelve independent current generators nor a nonzero bracket, and only the identity proper recharting lies in this word algebra. This obstruction is limited to that source packet. A source construction with port-indexed reversible perturbations and both composition orders could carry the missing antisymmetric response and closed overlap words.

The target-free diagonal phase lift supplies twelve independent first derivatives and an abelian bracket. Adjoining the registered connected adjacency tangent generates $\mathfrak{u}(12)$, whose derived algebra has dimension 143. The required response therefore needs a non-diagonal source law with derived rank 11, rather than either of these direct lifts. Conditional on the canonical oriented carrier, the complete target-free space $\operatorname{Hom}_{A_5}(\Lambda^2\mathbb{Q}^{12}, \mathbb{Q}^{12})$ has dimension 14, with an exact rational Reynolds basis. This fixes the search space without selecting a bracket. The complete Jacobi system has quadratic coefficient-row rank 38 and an exact $11 + 27$ rowspace split. Conditional on three named compact-Lie inputs, the real compact locus is classified into families P, F, G. A declared equal-weight oriented-face rule lands exactly on $60R_{13}$, fails Jacobi in 240 coordinates, and has G as its unique nearest compact family under the three audited coordinate norms, while the exact invariant-metric tie surface has a nonempty F region and excludes P on the complete invariant cone. The committed dual-measure theorem sharpens the comparison point: equal face weights and the barycentric one-third share give uniform port-dual weight $1/12$, whose diagonal metric is exactly the balanced cone point, where G is uniquely nearest with exact distances twelve times the reference values. The weighting rules, the measure-to-metric identification, the nearest-point law, and the metric class are not consequences of the source axioms. Noncompact components, source bracket selection, reconstruction, and same-current holonomy remain undischarged [9,10].

Trace balancing extends the same block construction to $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$. Its connected group is

$$S(U(3) \times U(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

The six-element kernel is exact for the declared tensor table. The independent axis calculation has Smith invariants $(1, 1, 1, 1, 1, 6)$ after diagonal and zero-sum axis relations are declared. The carrier measures the six axes and their deck action. It does not select those relations. Equality of the two group orders and an isomorphism between chosen generators therefore give a conditional algebraic match, not a physical global-form selection. A source-derived complete character category, a same-source loop-to-kernel identity, four-dimensional instanton normalization, and laboratory flux attachment are premises of that selection.

The trace-balanced block $V = \mathbb{C}^3 \oplus \mathbb{C}^2$, with hypercharges $(-1/3, 1/2)$, has the exterior branching

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L.$$

It contains fifteen states, the Standard Model hypercharges for one generation, the expected three one-Higgs invariant lines, cancelled gauge and mixed anomalies, and exactly four weak doublets. This is a precise representation witness. The finite response source does not select the charged-double-triplet current or this matter action. Laboratory matter identification, exclusion of extra light sectors, and attachment of three physical families require separate source receipts [3].

The screen coefficient module also gives an exact family-band selection under its named premises. Its four bands have ranks $1, 3, 3, 5$. Restricting to single complete faithful objects in the conditional window $3 \le N_g \le 5$, the Laplacian costs $5-\sqrt{5}, 6, 5+\sqrt{5}$ select the first rank-three band uniquely. The declared unitary simulator recovers that projector at its lowest positive generator frequency. Identifying this finite response mode with physical matter families requires matter-pole identification, continuum Spin/locality, physical seam selection, persistence, and laboratory receipts. Tensoring the rank-three band with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ data come from that table. A different receipt checks the declared local extension $D_\sigma \otimes I_{45}$ and conditional inheritance of the signed seam operator's positive finite-domain gap. This action is not source-selected. The twelve-port Spin packet has no certified bridge to the 8,662-node local domain. The finite gap is distinct from the compact-gauge repair spectrum and the continuum Yang–Mills mass gap.

The record-counting mechanism is operational on the same twelve-port architecture. A write appends $+1$, a retraction appends -1, and the readback sums these atomic events at each port. A conservative repair transfers one whole unit across a seam whenever its oriented mismatch has magnitude at least two. If $V(N) = \sum_i N_i^2$, a repair from the larger endpoint to the smaller one changes V by $-2(d-1)$, where $d \geq 2$ is the mismatch. The repair therefore terminates. Divisibility of the total load by twelve is a necessary consensus condition. An explicit eighteen-move witness settles the declared full-pile packet. The minimum admissible move count is natural under all sixty carrier rotations and the declared refinements. A half-unit display rescales the same event graph and threshold, so it changes units rather than the atomic mechanism or its cost.

The finite theorem forces the local Standard Model gauge Lie algebra. The separate transportable-sector route reconstructs a compact group, and the declared Standard Model sector packet has the same local type. Identity of the two current objects on one physical source is a separate premise.

5. Arithmetic closure, electroweak analytic checks, and charged-lepton diagnostics

The local pixel fixed point

The declared local map relates an outer detuning coordinate to an electromagnetic endpoint returned at the same trial input:

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

Here $A_T(P)$ is the inverse electromagnetic coupling returned by the declared Thomson-limit map. For each declared map, outward-rounded interval Banach certificates prove existence and local uniqueness. A 256-piece derivative enclosure proves global at-most-one across the analytic domain, with an empty exceptional set [5,10]. The gauge-width map has the certified root

$$\alpha^{-1} = 137.035660136946577\ldots$$

beside the 2022 Committee on Data of the International Science Council (CODATA) value 137.035999177(21) [11]. The relative separation is 2.5×10^{-6}.

Neither source fixed-point solve imports the measured Thomson value. Their physical interpretation requires the missing source-derived, same-scheme hadronic transport, target-independent map selection, and a typed bridge proving that both coordinates read the same physical quantity. A separate mixed diagnostic combines the source root with a coupling evaluated at a coordinate derived from the CODATA value. That quantity is a comparison, not a source-only prediction. The certified roots do not constitute an independent determination of the physical fine-structure constant.

Strict W/Z analytic checks

The source-side branch emits the electroweak chart coordinates

$$(M_W, M_Z) = (80.330,\ 91.119)\ \mathrm{GeV}.$$

Writing $S = gv_F/2$, $t = g'/g$, $w = S^2$, and $z = S^2(1+t^2)$, on the declared domain $S \neq 0$, $1+t^2 \neq 0$, and $1+d_Z \neq 0$, the strict one-loop consumer has the exact factorization

$$\frac{s_W}{s_Z} = \frac{1+d_W}{(1+t^2)(1+d_Z)},$$

where $d_W = \Delta_{WW}(w)/w$ and $d_Z = \Delta_{ZZ}(z)/z$. This is the exact quotient of the one-loop-truncated pole coordinates. Its strict one-loop expansion is $\frac{1}{1+t^2}(1 + d_W - d_Z) + O(\mathrm{loop}^2)$. The explicit common scale cancels under passive common-unit rescaling at fixed normalized self-energy factors; an active change of v_F can move d_W and d_Z through thresholds. This identity leaves t, d_W, d_Z to be supplied and is not a numerical postdiction. Separate contour arithmetic excludes zeros on the declared principal-sheet boxes. Separate receipts isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. These results identify neither chart with the physical resonance sheet and prove no unique continuation from the principal chart, sign bridge, full-matrix Laurent residue or physical current amplitude, or an independent numerical replay. The external self-energy fixture is not composed with the OPH coordinates above, so no chart-to-pole map or W/Z mass comparison follows.

The charged-lepton interval surface

The charged-lepton calculation combines a finite eight-path carrier with an empirical electromagnetic transport packet. Independent 100-digit decimal recomputation gives outward-rounded enclosures with logarithmic half-width 1.732%, corresponding to one-sided multiplicative widths of -1.72% and $+1.75\%$. The measured triple lies inside all three enclosures. A narrower conditional eight-register coordinate lies about 84 parts per million from the comparison triple [6,10].

Measured mass ratios and transport anchors enter this branch, so these numbers are diagnostics of internal closure. They give a sharp test surface without claiming a source-only mass prediction. A physical charged determinant line, an absolute clock, and a source-emitted transport bridge are required for an endpoint prediction.

An exact positive-chamber Koide theorem

Let $R^3 = I$ and consider the Hermitian three-cycle response

$$C = aI + bR + \bar{b}R^2.$$

When its three eigenvalues are nonnegative and are read as square-root masses, the normalized mass coordinate satisfies

$$Q = \frac{\sum_i m_i}{\left(\sum_i \sqrt{m_i}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{|b|}{a}\right)^2.$$

Consequently,

$$Q = \frac{2}{3} \iff \frac{|b|}{a} = \frac{1}{\sqrt{2}}.$$

The phase cancels from Q and jointly controls the two mass ratios. Equal rank-two event blocks in the finite tracial Gelfand–Naimark–Segal packet give the required modulus balance exactly [7,10]. Lean checks the circulant identity and equivalence. No physical chiral-family attachment or phase selection is claimed, so the theorem is independent of the charged-lepton diagnostic above.

Under the balance and mass-ordering premises, the measured electron and muon masses determine the tau mass through one quadratic. The outward-rounded enclosure is [1776.968991, 1776.969063] MeV, a 72-eV window sitting 0.43σ from the measured 1776.93 ± 0.09 MeV, with the spurious root excluded below the muon mass and the premise ancestry declared: the balance premise is target-informed by the measured triple, so this is a conditional postdiction. The registered test has a precision floor of 0.045 MeV. For a qualifying average, a central value more than three reported standard uncertainties from 1776.969027 MeV refutes the balanced-circulant premise; compatibility requires agreement within two standard uncertainties. Every other result is inconclusive. The target definition, kill band, decision rule, and calendar custody are fixed [13].

A frozen primitive-port propagation prediction

Let u_i be the twelve unit carrier directions. The physical branch uses a real, reciprocal, finite-range cosine kinetic symbol. The complete primitive orbit is its sole hop support through sixth derivative order, with no independent isotropic or directional term through that order. Proper carrier covariance makes every port weight equal. Normalizing the quadratic continuum term fixes

$$\omega^2(k,n) = \frac{1}{2a^2}\sum_{i=1}^{12}\left[1-\cos(ak\,u_i\cdot n)\right].$$

Its exact long-wavelength expansion is

$$\omega^2 = k^2 - \frac{a^2}{20}k^4 + \frac{a^4}{840}k^6 + \frac{2a^4}{7875}k^6 I_6(n) + O(a^6k^8),$$

where $I_6 = (25/132)\sum_i P_6(u_i\cdot n)$ is normalized to one on a vertex direction. “Spin six” means spherical-harmonic rank six, unrelated to particle spin. The coefficient vector obeys

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}.$$

For the declared equal-weight cosine kernel, put $x = |ak|$. On $0 < x \le 1$, its angular part through eighth order is

$$A(x)I_6(n), \qquad A(x) = \frac{2x^6(30-x^2)}{118125} > 0.$$

After division by $A(x)$, the complete x^{10}-and-higher tail E_x obeys the exact bounds

$$\|\nabla E_x\| \le \frac{6875}{101152} < \frac{7}{100}, \qquad \|\operatorname{Hess}_{S^2} E_x\| \le \frac{383125}{562658} < \frac{7}{10}.$$

The 62 critical directions of I_6 form 31 projective axes with exact minimum separation $\sin^2\theta = 1/2 - \sqrt{5}/6$. An exact three-chart interval cover and local singular-value bounds prove that the full kernel has exactly these 62 stationary directions for every $0 < x \le 1$, with twelve maxima, twenty minima, thirty saddles, and the same Morse indices [9]. These relations belong to the selected vertex-orbit branch. The certified repair dynamics acts on thirty internal seams. The source boundary and signed load readout map those seams onto the even-sum lattice $D_6 \subset \mathbb{Z}^6$. With the pullback of the response-selected Gram metric, this lattice has the same three-dimensional completion as the signed source-load module. The usual six-dimensional lattice metric is not used. Each directed seam maps to a carrier difference, and the normalized complete direction multiset is the thirty-direction edge orbit.

Cumulative record addition supplies an exact homogeneous internal action. Under explicit A2-natural feasibility and objective premises plus an A3 unique-minimizer premise, every directed seam has weight 1/60. The resulting Markov average P preserves constants and positivity and is reversible for counting measure. Its generator $L = I - P$ satisfies the positive maximum principle, is diagonalized by plane waves, and obeys

$$2fLf - L(f^2) = \frac{1}{60}\sum_e \big(f(x+v_e) - f(x)\big)^2 \ge 0.$$

A seam event has squared norm 4 in the raw moment chart and $a_{\text{edge}}^2 = 2 - 2/\sqrt{5}$ in the unit-diagonal response completion. The latter is an exact internal dimensionless scale rather than a physical length. The exact internal character is

$$\Lambda_{\text{int}}(\mathbf{k}) := \frac{6}{a_{\text{edge}}^2}\lambda_L(\mathbf{k}).$$

The algebraic phase lift $(q,p) \mapsto (p,-Lq)$ obeys $q'' + Lq = 0$ and $\omega_{\text{aux}}^2 = \lambda_L$. Its second component and spectral root are auxiliary algebraic objects; they construct no trajectory, clock, energy, or physical field equation. A physical dilation that maps the internal step to a length a would instead give

$$\Lambda_a(k,n) = \frac{6}{a^2}\lambda_L\left(\frac{a}{a_{\text{edge}}}kn\right),$$

where $\mathbf{k} = kn$, $k = |\mathbf{k}|$, and $|n| = 1$. The dilation and the identification $\Lambda_a = \omega_{\text{phys}}^2$ are separate physical premises.

If these currents act as one homogeneous physical position law, form the sole equal-weight support of one scalar or polarization-independent sector, glue across charts and refinement, and share one physical scale and frame, the spatial symbol is

$$\Lambda_a(k,n) = \frac{1}{5a^2}\sum_{j=1}^{30}\left[1-\cos(ak\,w_j\cdot n)\right].$$

Its exact coefficients are

$$C_4 = -\frac{a^2}{20}, \qquad B_0 = \frac{a^4}{840}, \qquad B_6 = -\frac{a^4}{12600},$$

with $B_0/C_4^2 = 10/21$, $B_6/C_4^2 = -2/63$, and $B_6/B_0 = -1/15$. Writing $q = ak$, define

$$\widehat{\Lambda}(q,n) = \frac{1}{5}\sum_{e=1}^{30}[1-\cos(q\,w_e\cdot n)].$$

The exact eighth moment is $M_8 = 10/3 - (8/15)I_6(n)$. For

$$P_6 = q^2 - \frac{q^4}{20} + \left(\frac{1}{840} - \frac{I_6}{12600}\right)q^6, \quad P_8 = P_6 + \left(-\frac{1}{60480} + \frac{I_6}{378000}\right)q^8,$$

alternating cosine bounds and $-5/9 \le I_6 \le 1$ give, for $0 \le q \le 1$,

$$|\widehat{\Lambda} - P_6| \le \frac{7}{388800}q^8, \quad 0 \le \widehat{\Lambda} - P_8 \le \frac{7}{34992000}q^{10}, \quad \frac{19}{20}q^2 \le \widehat{\Lambda} \le q^2.$$

This target-free result controls the finite spatial symbol. If Λ_a is identified with physical ω^2_{phys}, its positive formal branch gives

$$\omega = k - \frac{a^2}{40}k^3 + a^4\left(\frac{19}{67200} - \frac{I_6}{25200}\right)k^5 + O(a^6k^7),$$

and

$$\frac{\partial\omega}{\partial k} = 1 - \frac{3a^2}{40}k^2 + a^4\left(\frac{19}{13440} - \frac{I_6}{5040}\right)k^4 + O(a^6k^6).$$

The transverse group-velocity term is

$$\mathbf{v}_\perp := \frac{1}{k}\nabla_{S^2}\omega = -\frac{a^4k^4}{25200}\nabla_{S^2}I_6 + O(a^6k^6).$$

These series are derived without comparison data. They require an analytic remainder for the physical-frequency branch and physical position, field, clock, gluing, scale, frame, readout, and nuisance maps. The vertex and edge rays carry opposite rank-six signs and cannot be combined or selected after exposure. On a branch selected without using comparison data, a resolved negative C_4 fixes the carrier scale and that branch's two sixth-order amplitudes. The residual freedom is one three-parameter orientation class in $\mathrm{SO}(3)/A_5$. Intrinsic angular ranks one through five vanish, and binary refinement reduces B_6 by sixteen.

Both branch targets and their decision rules carry public commit custody. The detached timestamp evidence consists of calendar commitments and does not claim Bitcoin confirmation [13]. For the primitive-port branch, the dated cosmic-microwave-background template search and every data product examined in it form an excluded exposure class. The edge branch separately excludes that template class, every primitive-port comparison input, and every comparison datum inspected before its freeze. Neither linked coefficient manifold has an eligible physical comparison. A qualifying test requires one physical sector to realize the scalar symbol, equal action on both transverse polarizations for a photon test, a coherent carrier frame, and isolation of the intrinsic coefficient vector. If C_4 is negative at five or more standard deviations with enough sensitivity, exclusion of the linked B_0/B_6 terms or the rotated I_6 vector at five or more standard deviations, after the fixed $\mathrm{SO}(3)/A_5$ profile and with calibrated joint coverage, rejects this physical branch. An isolated positive C_4 or nonzero intrinsic anisotropy at ranks one through five also rejects the branch at the same threshold and under the same calibrated coverage. A calibrated joint likelihood that excludes the complete branch manifold at five or more standard deviations also rejects it. A null, underpowered, incomplete-covariance, unresolved-frame, polarization-split, or non-isolated result is inconclusive on the vertex branch. On the edge branch, a null verdict requires a pre-exposure same-action lower bound $a_{\min} > 0$ and a preregistered power calculation that makes the entire admitted $a \ge a_{\min}$ manifold excludable by the joint likelihood. Without both, a null is inconclusive. Support requires exclusion of the zero-coefficient baseline at five standard deviations or more, agreement with the complete linked branch within two, rejection of the named systematic alternatives, and an independent eligible replication. Minimal locally Lorentz-invariant Standard Model physics with General Relativity supplies the zero-coefficient baseline. Nonminimal effective operators can imitate the pattern, so a match favors the branch over that baseline without uniquely identifying OPH. This result does not derive the physical-sector bridge from the repair law. A failed test reaches OPH as a whole only if a derivation proves the branch forced and exclusive.

An exposed retrospective threshold translation gives a conditional scale diagnostic without a physical verdict. Under the unresolved photon-sector and frequency-squared identifications, conversion to the Pierre Auger energy-basis convention gives $\delta_{\gamma,2} = -a^2/20$ at leading order. The published calculation additionally assumes ordinary additive conservation in the cosmic-background frame, Lorentz-invariant electron and positron dispersion, and photon-sector Lorentz violation. On that same conditional branch, the exact symbol obeys $\Omega_\gamma(k,n) \le |k|$ globally for nonzero a. The Taylor-remainder certificate separately uses $0 \le ak \le 1$. Positive-mass leptons with Lorentz-invariant positive-energy dispersion therefore make decay into an electron-positron pair kinematically impossible. At fixed incoming momenta, the Lorentz-invariant incoming energy is no smaller than the seam-current incoming energy, so the seam-current energy-budget domain is contained in the Lorentz-invariant one. These conclusions use the full finite symbol rather than its effective-field-theory truncation and hold for every carrier direction.

Use the leading dispersion convention $E_i^2 = p_i^2 + m_i^2 + \delta_{i,2}E_i^4$, with $i \in \{\gamma,+,-\}$, $m_\gamma = 0$, $m_+ = m_- = m_e$, E the hard-photon energy, and ϵ the soft-photon energy. Taking the soft background photon to be Lorentz invariant at leading order, independent charged-lepton coefficients and a fixed positron energy share $0 < x < 1$ give the leading head-on, collinear threshold equation

$$\left[\delta_{\gamma,2} - x^3\delta_{+,2} - (1-x)^3\delta_{-,2}\right]E^4 + 4\epsilon E - \frac{m_e^2}{x(1-x)} = 0.$$

At equal sharing, only $\delta_{\gamma,2} - (\delta_{+,2} + \delta_{-,2})/8$ is visible. The coefficient map has a two-dimensional fiber, so a photon-only scale bound requires the published Lorentz-invariant-lepton premise or an independent lepton derivation. On that special branch, equal sharing uniquely minimizes the mass penalty and maximizes the leading head-on, collinear residual. Under the same branch, the alternative source scenario containing a subdominant proton component up to 10^{20} eV reports $\delta_{\gamma,2} > -10^{-58}\,\mathrm{eV}^{-2}$, which implies

$$a < 8.825 \times 10^{-36}\,\mathrm{m} = 0.546\,\ell_P.$$

The reference source scenarios yield no electromagnetic constraint, and the direct bound has no stated confidence level. The Auger calculation also neglects direct Lorentz-violating changes of the Breit–Wheeler cross section and depends on source, propagation, atmospheric-shower, detector-response, and frame attachments absent from the finite source theorem. The translation is a conditional upper bound on an open scale and carries no OPH verdict or evidence weight [9,11,14].

6. Fixed capacity and the de Sitter shock sign

For pure de Sitter space of radius L in spacetime dimension d,

$$r_c = L, \qquad \kappa = L^{-1}, \qquad \mu^2 = (d-2)\kappa r_c = d-2 = \lambda_{\ell=1}(S^{d-2}).$$

The radius and cosmological constant cancel. The smooth $\ell = 1$ modes are generated by de Sitter isometries and form the gauge kernel of the imported shock operator [8,13].

For finite sector dimensions d_i, total dimension $M = \sum_i d_i$, and sector probabilities p_i,

$$S_{\mathrm{gen}}(p,d) = -\sum_i p_i \log p_i + \sum_i p_i \log d_i = \log M - D_{\mathrm{rel}}(p \| d/M),$$

where D_{rel} is relative entropy. Thus $p_i = d_i/M$ gives the exact maximum $S_{\mathrm{gen}}^{\max} = \log M$. If an observer receives a fraction f of a fixed horizon-observer capacity and the horizon sectors deplete uniformly, every admissible integer transfer obeys

$$\Delta S_{\mathrm{gen}}^{\max} = \log(1-f) < 0.$$

The positive-real interpolation is strictly decreasing and concave. Its zero-transfer point is a one-sided boundary maximum.

The associated logarithmic sector observable has symmetric-point Hessian

$$\operatorname{Hess} \mathcal{A} = \frac{1}{nd^2}\left(I - \frac{2}{n}J\right).$$

Its fixed-total tangent curvature is positive, its homogeneous curvature is negative, and its symmetric-point gradient is nonzero. The one-sided capacity transfer carries the sign; an interior fixed-capacity extremum is excluded [8,10].

On the twelve-port icosahedral graph, the raw Laplacian spectrum and regular line-graph identity are exact. If the rotation triplet is an exact gauge sector and the physical shock kinetic term is the scaled nearest-neighbour port or edge Laplacian, normalization at that triplet gives

$$\{-2,\ 0,\ 1+3/\sqrt{5},\ 1+\sqrt{5}\}.$$

The edge carrier adds only eigenvalue 10 with multiplicity 18. Identifying capacity with horizon area, the transfer coordinate with observer mass, and the finite operator with the gravitational shock supplies a conditional time-advance mechanism. The finite identities make none of those physical identifications. They supply no positive cyclic static-patch trace. The obstruction of Chen, Stanford, Tang, and Yang excludes that trace proposal rather than the finite identities above [12].

7. Why the cumulative case is worth testing

Each receipt has a separate mathematical domain. OPH becomes interesting because the receipts share one finite architecture. Public records lead to the event algebra. Signed cumulative records and the normalized repair-response limit lead to an abstract continuous local three-dimensional Euclidean carrier. Spherical support leads to Lorentz kinematics. Modular and entropy data lead to the Einstein composition. Icosahedral carrier data lead to the gauge coefficient algebra and an exact matter witness. The same finite readback idea produces the pixel closure, the tracial Koide balance, and the capacity transfer law.

The dependencies are visible enough to be attacked. A proposed carrier must emit local state, ports or boundaries, readback, records, repair moves, and a public evidence bundle. Physical claims require source-bound maps between the finite objects and their spacetime, current, matter, clock, or horizon interpretations. Lean checks symbolic implications. Exact programs check finite searches and interval arithmetic. Simulations measure branches whose continuum construction lies beyond the finite theorem.

The falsification instrument uses target definitions, premise ancestry, cryptographic custody, kill bands, precision floors, and decision rules fixed before comparison [13]. The tau window has a custody-bound precision floor and decision rule. The Higgs envelope is a target-conditioned comparison without a predictive verdict. The charged-lepton anchor-gap band is a confirm-or-refute condition for a source-emitted bridge value; the source derivation of that value is a premise of the comparison. Landing inside confirms the decomposition; landing outside refutes it.

Conditional hypotheses that fail their predeclared comparison rules are excluded from the evidentiary case in this paper. The rejected hypotheses contribute nothing to the exact record, geometry, gauge, Koide, or capacity results [13].

The program therefore has a clear scientific shape. It contains exact mathematics with recognizable physical structure, a measured finite Lorentzian-signature signal with mechanism controls, certified arithmetic closures, strict electroweak scalar analytic checks, and explicit physical attachments that can succeed or fail. That combination supports a serious observer-first route toward unification. It does not amount to a completed physical theory of everything.

Assessment
OPH has a machine-checked core, exact finite constructions that land on the algebraic skeleton of known physics, and a reproducible support-sensitive Lorentzian-signature signal on a fixed chart. Its strongest physical conclusions depend on named carrier maps. The combination makes OPH a promising observer-first research program rather than a completed physical theory.

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AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-5.6-Sol, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.